Aether

# Data Analysis Report

Empirical diagnostic evaluation, descriptive statistics, and multivariate exploratory analysis.

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|---|---|
| DATASET NAME | **patients.csv** |
| REPORT DATE | 10 October 2026, 05:06 UTC |
| DATASET VERSION | Version 1 |
| TOTAL RECORDS | 5 rows |
| FEATURE DIMENSIONS | 5 columns (3 numeric, 1 categorical) |
| DATA QUALITY SCORE | **99/100** |
| SCOPE OF ANALYSIS | Full-Spectrum Exploratory & Statistical Audit |

**Notice:** This report is generated dynamically from verified local dataset records. All numerical metrics, distribution profiles, and correlation coefficients are computed strictly from the active dataset version.

## Executive Summary

| TOTAL RECORDS | TOTAL DIMENSIONS | DATA QUALITY SCORE |
|---|---|---|
| **5** | **5** | **99/100** |
| MISSING DATA RATE | DUPLICATE RECORDS | DETECTED OUTLIERS |
| **4.0%** | **0** | **0** |

This report presents a structured analysis for **patients.csv** (Version 1). The dataset comprises **5 observations** across **5 attributes**. Empirical evaluation yields an overall Data Quality Score of **99/100**, measuring record completeness (96%), uniqueness (100%), and validity (100%).

### Key Summary Findings:

• The dataset contains 5 verified observations across 5 attributes with an overall Data Quality Score of 99/100.

• Data completeness is 96%, with 1 missing values concentrated in ['blood_pressure'].

• Strongest linear association exists between 'blood_pressure' and 'cholesterol' (r = 1.00, strong positive).

• Categorical concentration observed in 'department' where 'Cardiology' accounts for 60.0% of total entries.

## Dataset Overview

| DATASET NAME | **patients.csv** | ANALYSIS DATE | 10 October 2026, 05:06 UTC |
|---|---|---|---|
| ROW COUNT | 5 | COLUMN COUNT | 5 |
| FILE FORMAT | CSV | MEMORY FOOTPRINT | 0.8 KB |

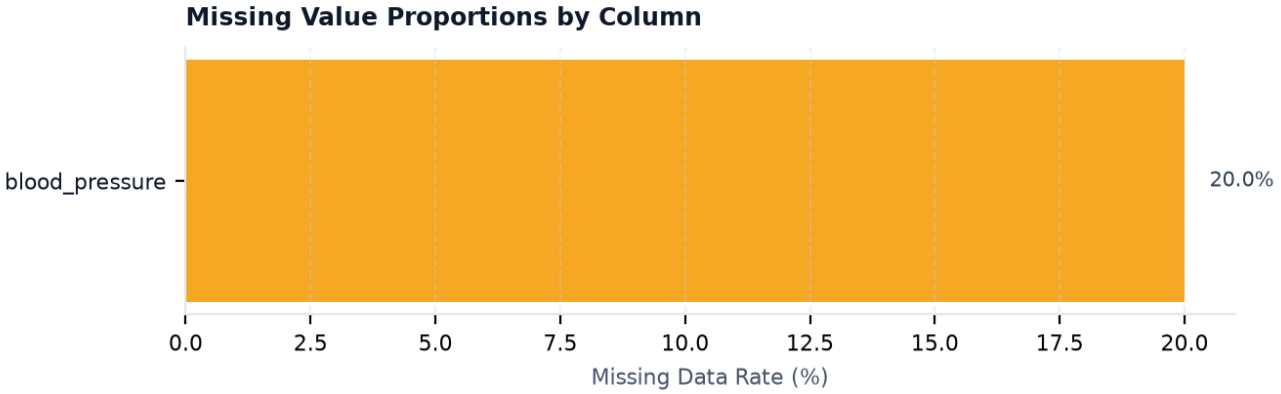
### Column Overview:

| Column | Data Type | Non-null | Missing | Unique | Summary |
|---|---|---|---|---|---|
| **patient_id** | str | 5 | 0 (0.0%) | 5 | 5 unique identifiers |
| **age** | int64 | 5 | 0 (0.0%) | 5 | [30.0 to 70.0], Med: 55.0 |
| **blood_pressure** | float64 | 4 | 1 (20.0%) | 4 | [110.0 to 150.0], Med: 130.0 |
| **cholesterol** | int64 | 5 | 0 (0.0%) | 5 | [180.0 to 290.0], Med: 240.0 |
| **department** | str | 5 | 0 (0.0%) | 3 | Top: Cardiology (60%) |

## Data Quality

| Quality Dimension | Score | Benchmark | Assessment Detail |
|---|---|---|---|
| **Completeness** | **96%** | >= 95.0% | Calculates proportion of non-null cells across the observation matrix. |
| **Uniqueness** | **100%** | >= 98.0% | Measures absence of identical duplicate row vectors. |
| **Validity** | **100%** | >= 90.0% | Evaluates presence of constant zero-variance features. |
| **Consistency** | **100%** | >= 85.0% | Checks for type homogeneity and uniform data formatting. |

**Audit Details:** 1 missing cells identified (4.0% of dataset). Duplicate row analysis identified 0 duplicate record(s) (0.0%).

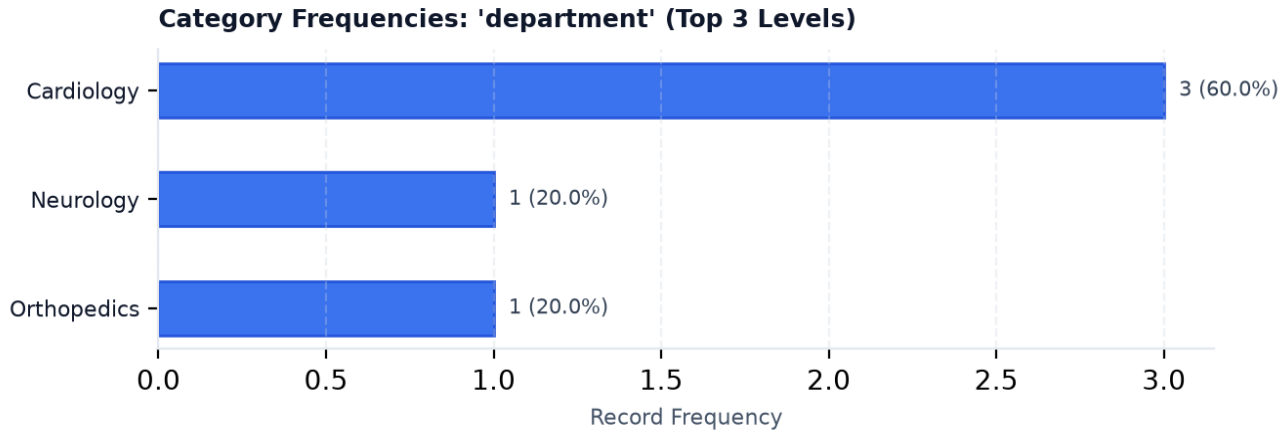



## Descriptive Statistics

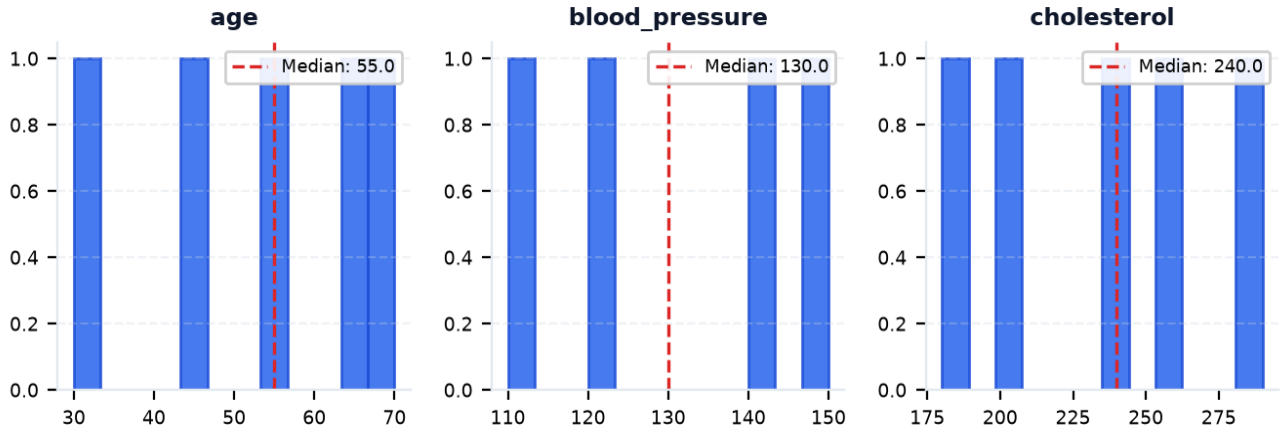
| Feature | Count | Mean | Median | Std Dev | Min | Max | Q1 | Q3 | IQR | Skew |
|---|---|---|---|---|---|---|---|---|---|---|
| **age** | 5 | 53.0 | 55.0 | 16.1 | 30.0 | 70.0 | 45.0 | 65.0 | 20.0 | -0.61 |
| **blood_pressure** | 4 | 130.0 | 130.0 | 18.3 | 110.0 | 150.0 | 117.5 | 142.5 | 25.0 | 0.00 |
| **cholesterol** | 5 | 234.0 | 240.0 | 44.5 | 180.0 | 290.0 | 200.0 | 260.0 | 60.0 | -0.02 |

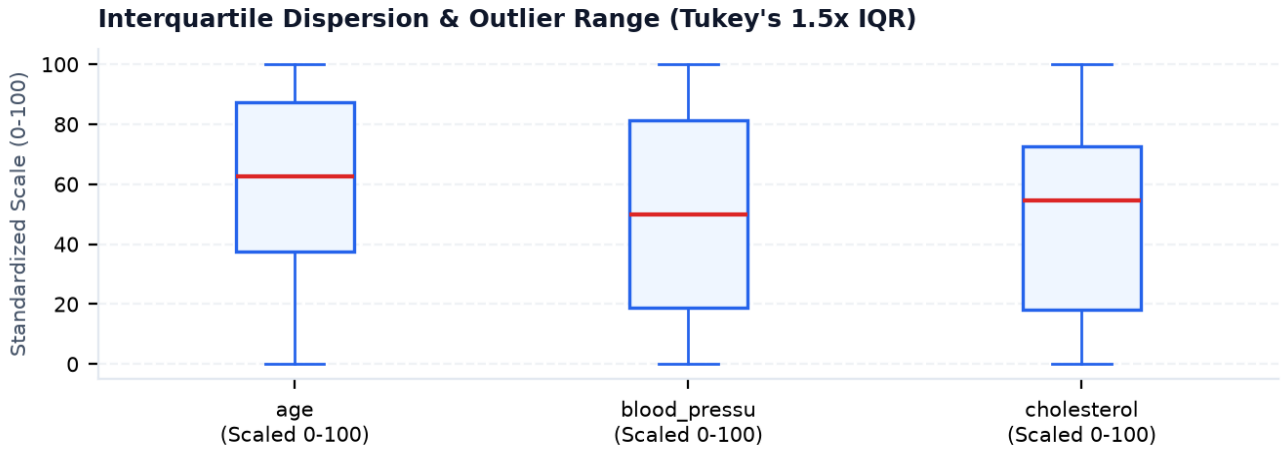
## Categorical Analysis

| Column | Unique | Most Frequent Value | Frequency | Percentage | Status |
|---|---|---|---|---|---|
| **department** | 3 | Cardiology | 3 | 60.0% | Dominant (60.0%) |

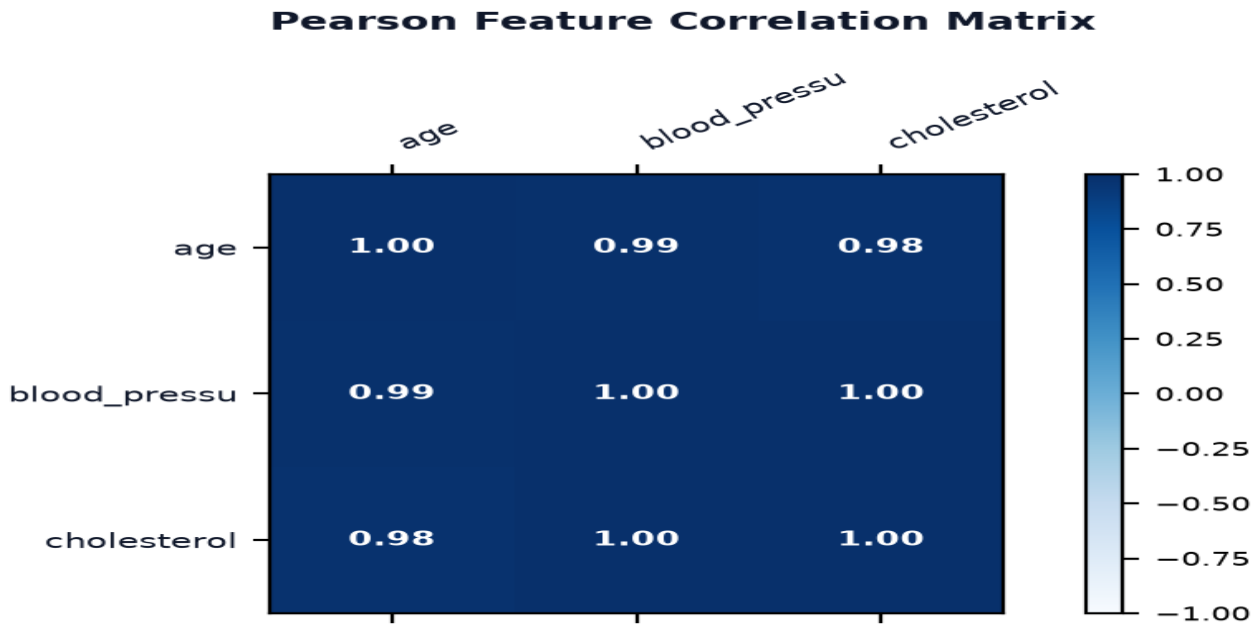



## Distribution Analysis



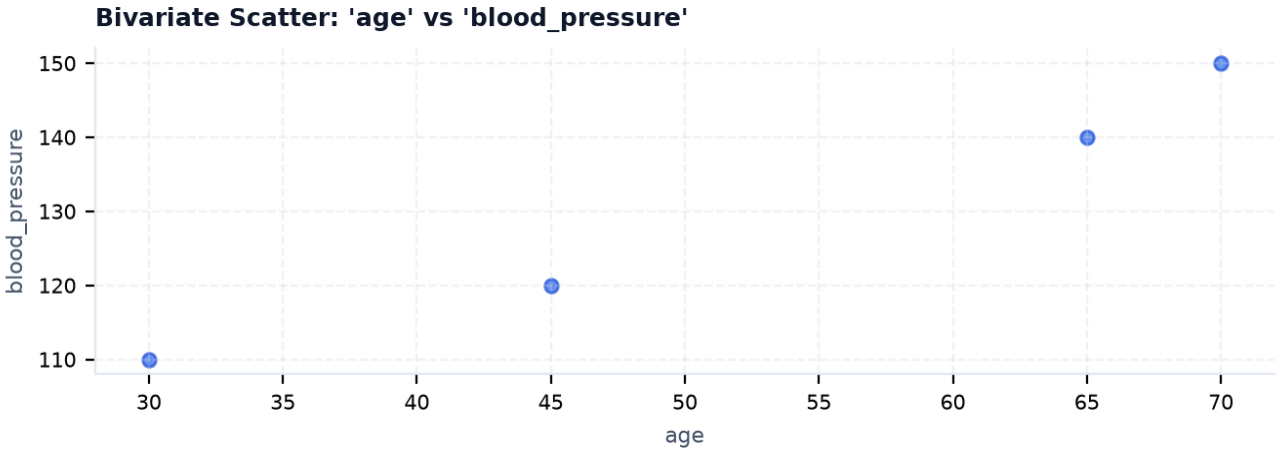



## Correlation Analysis




| Feature X | Feature Y | Pearson r | Strength | Direction |
|---|---|---|---|---|
| **blood_pressure** | **cholesterol** | 1.00 | Strong positive | Positive |
| **age** | **blood_pressure** | 0.99 | Strong positive | Positive |
| **age** | **cholesterol** | 0.98 | Strong positive | Positive |

## Relationships




## Outlier Analysis

| Feature | Outlier Count | Percentage | Lower Bound (1.5x IQR) | Upper Bound (1.5x IQR) |
|---|---|---|---|---|
| **age** | 0 | 0.0% | 15.00 | 95.00 |
| **blood_pressure** | 0 | 0.0% | 80.00 | 180.00 |
| **cholesterol** | 0 | 0.0% | 110.00 | 350.00 |

No statistical outliers detected outside standard 1.5x IQR thresholds.

## Key Findings

• The dataset contains 5 verified observations across 5 attributes with an overall Data Quality Score of 99/100.

• Data completeness is 96%, with 1 missing values concentrated in ['blood_pressure'].

• Strongest linear association exists between 'blood_pressure' and 'cholesterol' (r = 1.00, strong positive).

• Categorical concentration observed in 'department' where 'Cardiology' accounts for 60.0% of total entries.

## Recommendations

| Category | Priority | Action Recommendation |
|---|---|---|
| **Operational** | **Low** | Monitor the dominant category 'Cardiology' in 'department' to identify potential sampling bias or operational dependency. Further domain validation is recommended. |
| **Analytical** | **Medium** | Evaluate 'blood_pressure' and 'cholesterol' for potential interaction terms or collinearity redundancy in multivariate regression. |
| **Data Quality** | **Low** | Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings. |

## Appendix

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|---|---|
| **Mathematical Methodology** | • **Tukey's IQR Outlier Rule:** Outliers identified strictly outside [Q1 - 1.5*IQR, Q3 + 1.5*IQR].<br>• **Pearson Correlation:** Linear association computed as r = cov(X, Y) / (std(X) * std(Y)).<br>• **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) measuring distribution symmetry. |
| **Data Quality Scoring** | Overall score computed via weighted evaluation: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale directly with missing cell frequency and duplicate rows. |
| **Compilation Provenance** | Generated by Aether Engine • ReportLab PDF Architecture • Python 3.0.6 / NumPy 2.5.3. Timestamp: 10 October 2026, 05:06 UTC. |